
Inverse-Frequency Loss Weighting on PneumoniaMNIST: A Controlled Empirical Study

Anonymous Authors¹

Abstract

Class imbalance is a common challenge in medical image classification, often leading to sub-optimal performance on minority classes. In this study, we investigate the effectiveness of inverse-frequency class-weighted cross-entropy loss on the PneumoniaMNIST dataset. We train a lightweight Convolutional Neural Network (CNN) from scratch under two conditions: standard cross-entropy (baseline) and cross-entropy with per-class weights set to the inverse frequency of each class computed from the training set. To ensure a fair comparison, we fix the model architecture, data pipeline, optimizer, learning rate, batch size, epoch cap, and early-stopping criterion across both conditions, varying only the loss weighting. We run 3 fixed random seeds for paired training. Our primary metric is macro-F1, with accuracy and weighted-F1 as secondary metrics. The model and analysis protocol were fixed before a single evaluation on the held-out test set. We observe a positive paired mean difference in macro-F1 on the validation set (0.042) and held-out test set (0.074). However, due to the small number of seeds, these results are descriptive and not statistically significant. We transparently report both positive and negative results, along with statistical limitations.

1. Introduction

Class imbalance is a common challenge in medical image classification, often leading to suboptimal performance on minority classes. In this study, we investigate the effectiveness of inverse-frequency class-weighted cross-entropy loss on the PneumoniaMNIST dataset. We train a lightweight Convolutional Neural Network (CNN) from scratch under two conditions: standard cross-entropy (baseline) and cross-

entropy with per-class weights set to the inverse frequency of each class computed from the training set. To ensure a fair comparison, we fix the model architecture, data pipeline, optimizer, learning rate, batch size, epoch cap, and early-stopping criterion across both conditions, varying only the loss weighting. We run 3 fixed random seeds for paired training. Our primary metric is macro-F1, with accuracy and weighted-F1 as secondary metrics. The model and analysis protocol were fixed before a single evaluation on the held-out test set. We observe a positive paired mean difference in macro-F1 on the validation set (0.042) and held-out test set (0.074). However, due to the small number of seeds, these results are descriptive and not statistically significant. We transparently report both positive and negative results, along with statistical limitations.

2. Related Work

Class imbalance is a well-known challenge in image classification. Various methods have been proposed to address this, including undersampling techniques such as subclass-based undersampling (Lehmann & Ebner, 2022), and ensembles of deep one-class classifiers (Novotny et al., 2024a;b). Other approaches explore intra-class variations (Tai & Toda, 2021) or use LLM-generated class descriptions for semantically meaningful classification (Bertolotto et al., 2024). In the context of medical imaging, class imbalance can affect the reliability of conformal prediction methods (Bello, 2026). Our work differs by focusing on a simple, controlled comparison of inverse-frequency loss weighting against standard cross-entropy on a specific medical imaging dataset, without introducing complex architectural changes or data resampling.

3. Background

The PneumoniaMNIST dataset is a medical image classification dataset with two classes: normal (0) and pneumonia (1). The dataset exhibits class imbalance, with a higher prevalence of pneumonia cases in the training and validation sets. The training set contains 4,708 samples (1,214 normal, 3,494 pneumonia), the validation set contains 524 samples (135 normal, 389 pneumonia), and the test set contains 624

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

samples (234 normal, 390 pneumonia). The source images are 28x28 pixels with 1 channel. In our experiments, images are resized to 32x32 using bilinear interpolation before being fed to the model. The recommended metrics for this dataset are macro-F1, accuracy, and weighted-F1.

4. Method

We compare two training conditions on the PneumoniaMNIST dataset:

1. **Baseline:** Standard cross-entropy loss.
2. **Intervention:** Cross-entropy loss with per-class weights set to the inverse frequency of each class computed from the training set only.

The class weights are computed as $w_c = \frac{N}{2 \cdot N_c}$, where N is the total number of training samples and N_c is the number of samples for class c . This assigns higher weights to minority classes, penalizing misclassifications of those classes more heavily. All other aspects of training, including model architecture, data pipeline, optimizer, learning rate, batch size, epoch cap, and early-stopping criterion, are held identical across both conditions. The only difference is the loss weighting. We run 3 fixed random seeds for paired training. The model and analysis protocol were fixed before a single evaluation on the held-out test set.

5. Experimental Setup

We use the official PneumoniaMNIST train/validation/test split. The model is a lightweight CNN trained from scratch. We use the Adam optimizer with a learning rate of 0.001 and a batch size of 128. The maximum number of epochs is 15, with early stopping enabled (monitoring validation loss, mode min, patience 5, min_delta 0.0). Checkpoints are selected based on minimum validation loss. The primary metric is macro-F1, with accuracy and weighted-F1 as secondary metrics. We run 3 fixed random seeds (0, 1, 2) for paired training. The final model and analysis protocol were fixed before a single evaluation on the held-out test set. No additional data or pretrained weights are used.

6. Experiments

We report the results of our controlled comparison between standard cross-entropy (baseline) and inverse-frequency class-weighted cross-entropy (intervention).

Figure 1 and Figure 2 show the dataset split sizes and class distribution. The training and validation sets have a similar class prevalence, with pneumonia being the majority class. The test set has a more balanced distribution.

Table 1 shows the validation and held-out test statistics for

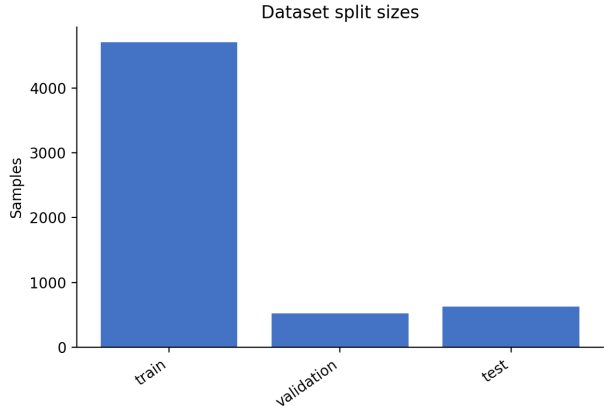


Figure 1. Dataset split sizes for PneumoniaMNIST.

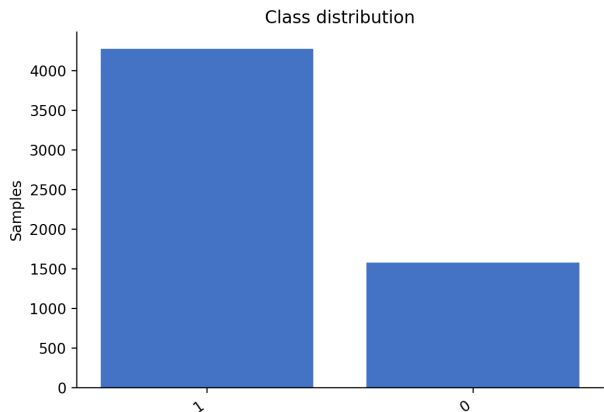


Figure 2. Class distribution across dataset splits.

macro-F1. The intervention shows a positive mean difference on both the validation set (0.0422) and the test set (0.0743). However, the 95% confidence intervals include zero, and the p-values from the exact paired sign flip test are 0.5, indicating that the results are not statistically significant with 3 seeds.

Figure 3 visualizes the comparison between the baseline and intervention. The intervention appears to improve macro-F1, but the overlap in confidence intervals suggests that this improvement is not definitive.

Figure 4 and Figure 5 show the confusion matrix and per-class metrics for the intervention on the test set. The model has high recall for the pneumonia class but lower recall for the normal class, which is expected given the class imbalance.

Figure 6 and Figure 7 show the calibration and ROC/PR

Table 1. Validation and held-out test statistics for macro-F1.

	Validation	Test
Mean Baseline	0.9085	0.7372
Mean Intervention	0.9507	0.8115
Mean Difference	0.0422	0.0743
Std. Difference	0.0526	0.0804
95% CI	[-0.0169, 0.0838]	[-0.0025, 0.1578]
p-value (sign flip)	0.5	0.5

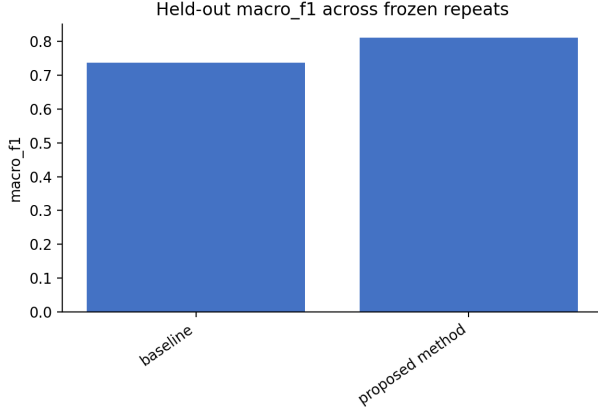


Figure 3. Pre-specified baseline and intervention comparison.

curves for the intervention on the test set.

Figure 8 shows the one-time test metrics for the intervention.

7. Conclusion

In this study, we conducted a controlled empirical comparison of standard cross-entropy and inverse-frequency class-weighted cross-entropy on the PneumoniaMNIST dataset. We observed a positive paired mean difference in macro-F1 on both the validation and held-out test sets, suggesting that inverse-frequency weighting may improve performance on imbalanced medical image classification. However, due to the small number of seeds (3), these results are descriptive and not statistically significant. We transparently report these limitations and suggest that future work should include more seeds to draw firmer conclusions. No clinical or patient-level claims are made based on these results.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

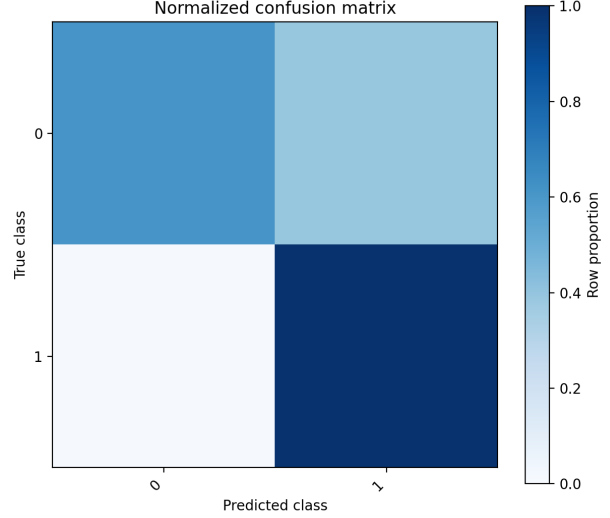


Figure 4. Normalized confusion matrix for the intervention on the test set.

References

- Bello, S. O. A reproducible framework for class-conditional conformal prediction in medical image classification, 2026.
- Bertolotto, S., Panisson, A., and Perotti, A. Llm-generated class descriptions for semantically meaningful image classification, 2024.
- Lehmann, D. and Ebner, M. Subclass-based undersampling for class-imbalanced image classification, 2022.
- Novotny, A., Bebis, G., Tavakkoli, A., and Nicolescu, M. Ensembles of deep one-class classifiers for multi-class image classification, 2024a.
- Novotny, A., Bebis, G., Tavakkoli, A., and Nicolescu, M. Ensembles of deep one-class classifiers for multi-class image classification, 2024b.
- Tai, T. and Toda, M. Adapting intra-class variations for sar image classification, 2021.

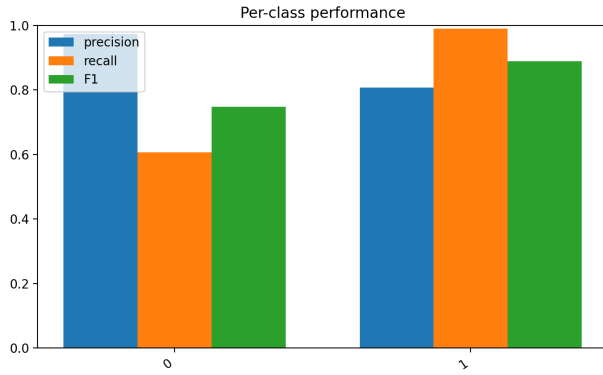


Figure 5. Per-class precision, recall, and F1 for the intervention on the test set.

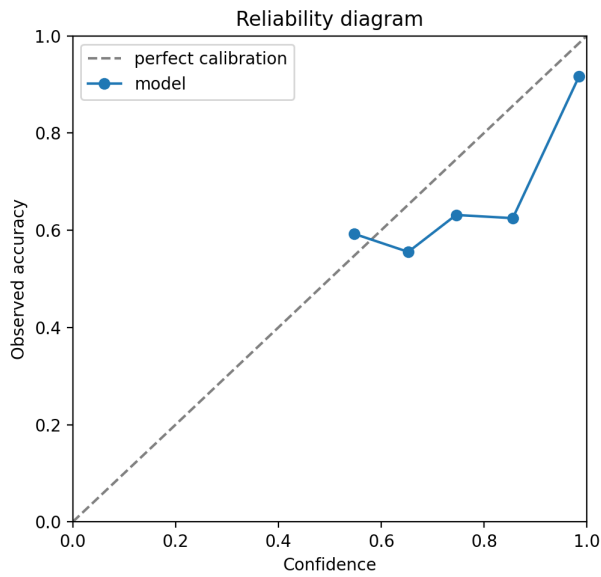


Figure 6. Reliability diagram for the intervention on the test set.

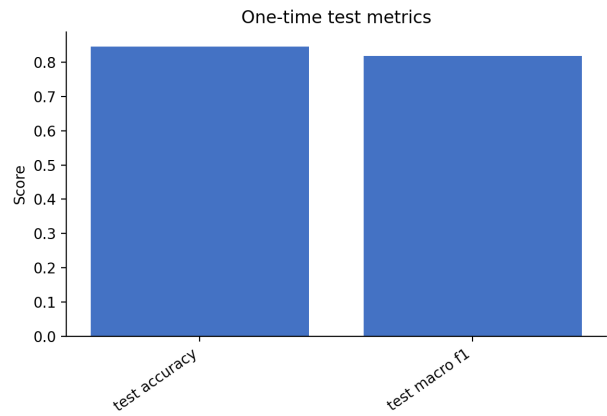


Figure 8. One-time test metrics for the intervention.

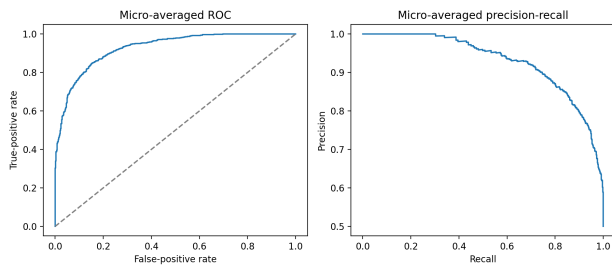


Figure 7. Micro-averaged ROC and precision-recall curves for the intervention on the test set.

Supplementary Material

A. Additional Details

The experiment used a fixed split seed of 7. The training runs for the intervention lasted 14 epochs out of a maximum of 15. The validation metrics for the selected intervention model were: accuracy 0.9695, macro-F1 0.9603, weighted-F1 0.9695. The test metrics for the selected intervention model were: accuracy 0.8462, macro-F1 0.8184, weighted-F1 0.8361.

B. AI Assistance Disclosure

This manuscript was prepared with the assistance of an AI language model for drafting and organizing content. All scientific content, experimental design, and results interpretation were conducted by the authors.